



UNIVERSITY OF KELANIYA - SRI LANKA
FACULTY OF SCIENCE

Bachelor of Science Honours Degree Examination, September/October - 2024

Academic Year: 2022/2023 - Semester I

ELECTRONICS AND COMPUTER SCIENCE

BECS 21413/BECS 21413 (R) - Analogue Electronics II (Operational Amplifiers)

No. of Questions: Seven (07) No. of Pages: Five (05) Time: 02 hours & 30 minutes

Answer any **FIVE (05)** questions.

(All the symbols and abbreviations have their usual meanings)

- 1) (a) Describe the op-amp's negative feedback concept using a suitable diagram. (03 Marks)
- (b) Using suitable circuit diagrams, derive expressions for the output voltage (V_o) of op-amp's basic inverting and non-inverting closed-loop configurations. Using the derived expressions, obtain closed-loop gains for inverting and non-inverting configurations. (08 Marks)
- (c) Using suitable circuit diagrams and the expressions you have obtained in part (b), discuss the construction of inverter and voltage follower circuits. (04 Marks)
- (d) By analysing the circuit shown in Figure 1, determine the output voltage of the circuit when the switch is connected to the positions 1, 2, and 3 separately. Assume that the output voltage saturation of the op-amp is $\pm 10.5 V$.

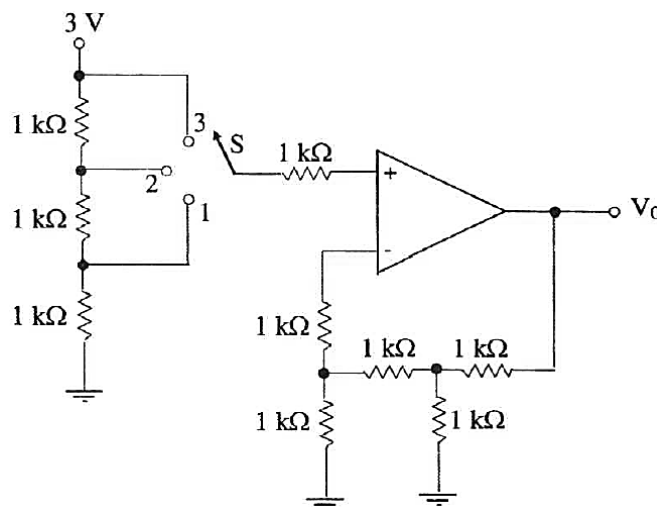


Figure 1

(10 Marks)

- 2) (a) The circuit diagram of a true differentiator is shown in Figure 2. Show that the output is the differentiation of the input function by deriving an expression for the circuit's output voltage. (05 Marks)

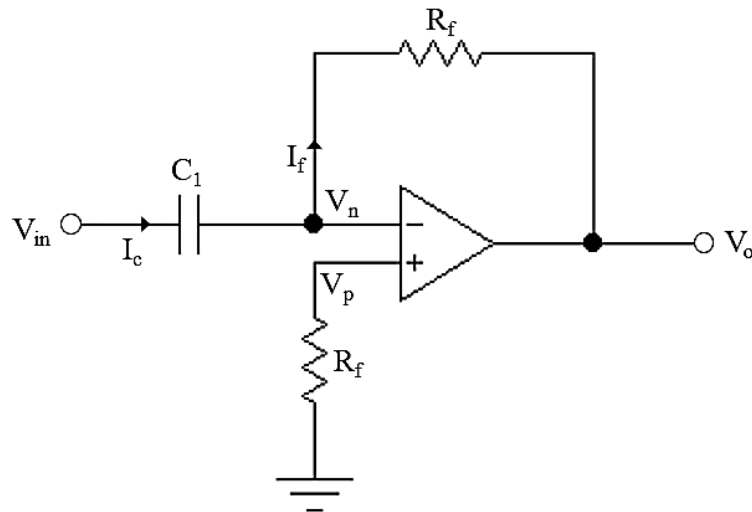


Figure 2

- (b) Derive the transfer function of the true differentiator and discuss its main design issue in response to very high-frequency input signals. (06 Marks)
- (c) Modify the above true differentiator to a practical differentiator circuit to mitigate the above design issue and derive its transfer function. (07 Marks)
- (d) Hence, show that the practical differentiator has a high-pass frequency response. (04 Marks)
- (e) What is the shape of the differentiator's output waveform you could observe when you input a square wave? Sketch both input and output waveforms on the same time scale. (03 Marks)
- 3) The circuit diagram of a phase shifter is shown in Figure 3.

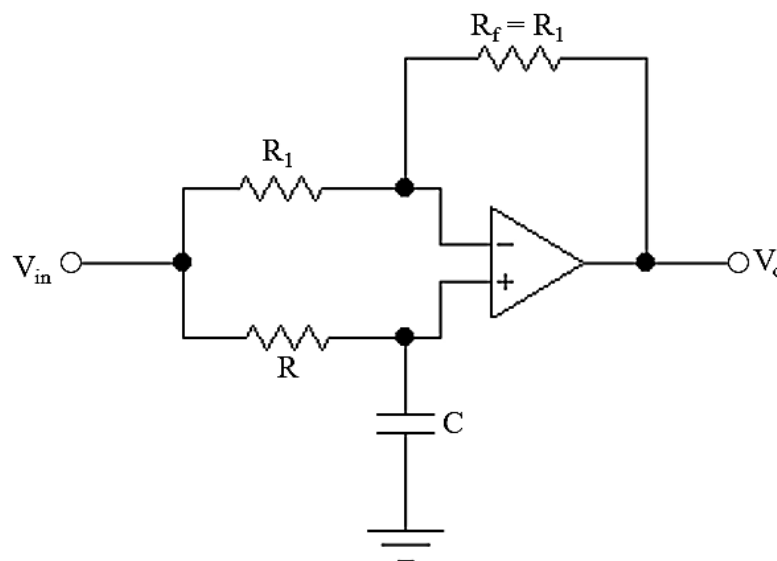


Figure 3

- (a) By applying the superposition theorem, show that the transfer function of the above phase shifter is given by,

$$T(jf) = \frac{1 - j\left(\frac{f}{f_c}\right)}{1 + j\left(\frac{f}{f_c}\right)} \quad ; \quad f_c = \frac{1}{2\pi RC} . \quad (10 \text{ Marks})$$

- (b) Determine the amplitude response of the phase shifter and, hence, explain how the phase shifter becomes an all-pass filter. (05 Marks)
- (c) Determine the phase response and state the type of the phase shifter. (05 Marks)
- (d) If $R_1 = 10 \text{ k}\Omega$, $R = 2 \text{ k}\Omega$ and $C = 0.01 \text{ }\mu\text{F}$, calculate the frequency of the input signal when the phase shift between input and output becomes 90° . (05 Marks)

- 4) (a) The circuit diagram of an inverting Schmitt trigger is shown in Figure 4. Obtain expressions for the upper and lower threshold voltages.

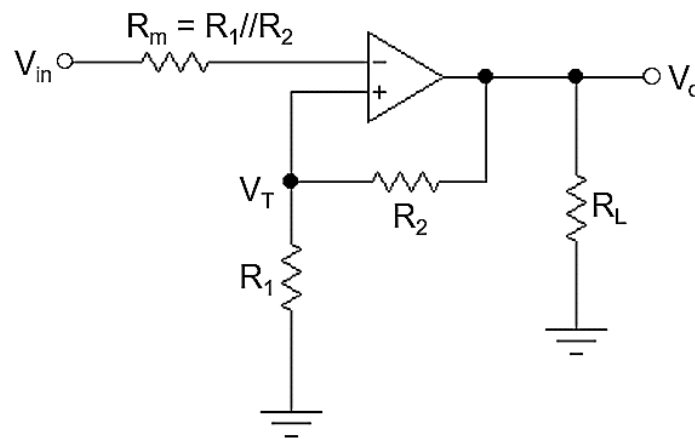


Figure 4 (04 Marks)

- (b) Explain the operation of the inverting Schmitt trigger using suitable diagrams and the voltage transfer curve. (05 Marks)
- (c) Modify the above Schmitt trigger to shift the trip points to the positive direction and obtain the new expressions for the upper and lower threshold voltages. Sketch the new voltage transfer curve. (07 Marks)
- (d) The temperature of an industrial oven has to be maintained between 200 and 240 Celsius using a heating element. The heating element should be automatically turned OFF when the oven temperature increases more than 240°C and turned ON when the oven temperature decreases less than 200°C . If a sensor is provided to identify the oven temperature, design a suitable circuit to control the temperature using an op-amp circuit. Note that the temperature sensor produces an analogue positive voltage that is proportional to the oven temperature. Assume that $V_{cc} = \pm 15 \text{ V}$ and $V_{sat} = \pm 13.5 \text{ V}$. (09 Marks)

- 5) (a) Draw the first-order RC passive low-pass and high-pass filter circuits and derive their transfer functions. Hence, obtain expressions for the cut-off frequency of both filters. (06 Marks)
- (b) Modify the above two circuits using an op-amp to have active gain control (non-inverting) and obtain the transfer functions for new circuits. (06 Marks)
- (c) Using the results of parts (a) and (b), design an active (non-inverting) band-pass filter with a **single-stage gain control** and obtain its transfer function. (08 Marks)
- (d) Design an active first-order band-pass filter having a pass band of 1 to 10 kHz and a gain of 2. (05 Marks)

- 6) Figure 5 shows the circuit diagram of a Wien-bridge network.

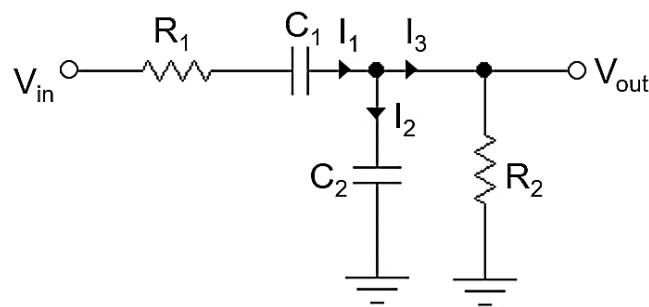


Figure 5

- (a) Derive the transfer function of the Wien-bridge network shown in Figure 5 and hence, obtain an expression for the resonant frequency when $R_1 = R_2 = R$ and $C_1 = C_2 = C$. (10 Marks)
- (b) Draw the Wien-bridge oscillator circuit using an op-amp and describe the conditions for oscillation. (08 Marks)
- (c) Design a Wien-bridge oscillator circuit having an oscillation frequency of 1 kHz. Use realistic values for all passive components. (07 Marks)
- 7) (a) Define the terms “Line regulation” and Load regulation” in relation to the voltage regulation. (06 Marks)
- (b) Draw the circuit diagram of a basic op-amp series regulator and explain its operation. (07 Marks)
- (c) Modify the series regulator in (b) with constant-current limiting and explain the operation of the current limiting circuit. (07 Marks)
- (d) The circuit diagram of a basic op-amp shunt regulator is shown in Figure 6.
- (i) In Figure 6, what power rating must R_1 have if the maximum input voltage is 12.5 V?
- (ii) Using the answer you obtain in (i), describe the shunt regulator's main drawback compared to the series regulator.

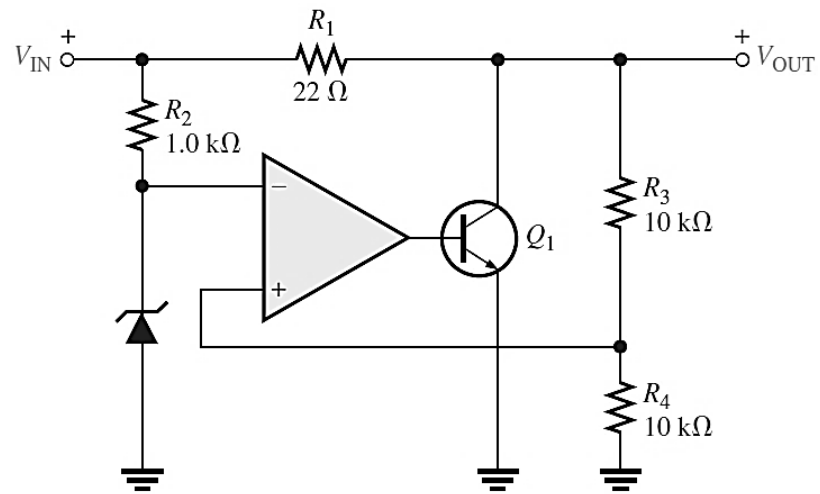


Figure 6

(05 Marks)
